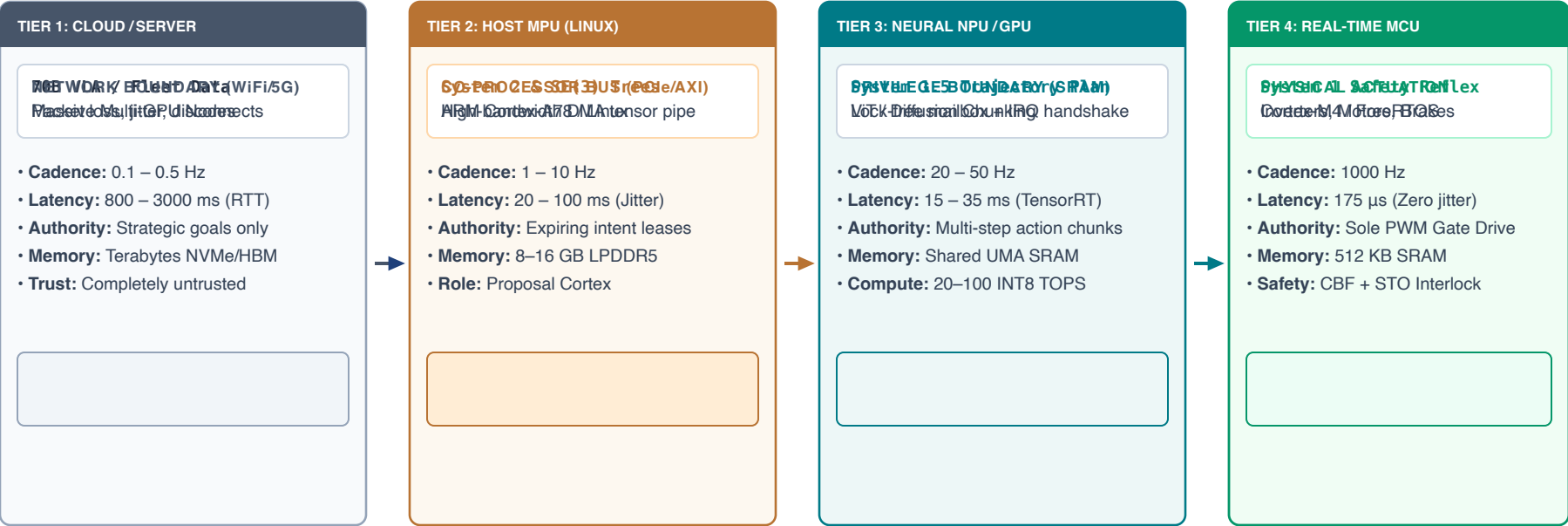


THE 4-TIER HETEROGENEOUS SILICON PLACEMENT MAP

Partitioning 7 agent organs across Cloud, Linux MPU, Neural Accelerator NPU, and Real-Time MCU tiers



HETEROGENEOUS PLACEMENT CODESIGN PRINCIPLES

1. **Physical Privilege Rule:** Only Tier 4 (Deterministic MCU) may hold physical motor PWM authority; Tiers 1–3 are untrusted proposal engines.
2. **Memory Hierarchy:** As rate increases (1 Hz $\rightarrow$ 50 Hz $\rightarrow$ 1000 Hz), memory transitions from high-capacity DRAM to lock-free zero-allocation SRAM.
3. **Panic Isolation:** Complete Linux MPU kernel crash or NPU bus hang leaves the 1 kHz MCU safety reflex executing deterministically.